

Appendix: Multitone Beating, Tone Phases, Pulse Area, and Diffraction Efficiency of the Crossed-AOD Profile

1 Notation

N_x, N_y	tones per axis	$\phi_{x,i}, \phi_{y,j}$	tone phases (free at the AWG)
W	tone span per axis	$a_{x,i}, a_{y,j}$	tone <i>power</i> weights
d	tone spacing, $f_n = f_{\text{off}} + nd$	$A_s = \sqrt{a_{x,i} a_{y,j}}$	spot field amplitude
f_0, T_0	fundamental beat freq., period	$u(r)$	focal mode, $u(0) = 1$
k_s	frequency order of spot s	ϱ	distance between two spots
H_m	group field of order m	θ	pulse area
D_δ	Fourier coefficient, beat order δ	η	diffraction efficiency
C	crest factor of the RF signal	α, P_s	peak efficiency, saturation power

Power versus voltage: r_x, r_y are *power* ratios, so

$$a = [\varrho, 1, \dots, 1, \varrho] \text{ in voltage, } [r, 1, \dots, 1, r] \text{ in power, } \varrho := \sqrt{r}. \quad (1)$$

2 Beating model

Spot frequencies and orders, with f_0 the greatest common divisor of the tone spacings:

$$f_s = f_{x,i} + f_{y,j} = f_{\min} + k_s f_0, \quad k_s \in \mathbb{N}_0, \quad f_0 = \frac{W}{\text{lcm}(N_x - 1, N_y - 1)}, \quad T_0 = \frac{1}{f_0}. \quad (2)$$

Spot field, spot phase, and the group field of order m :

$$g_s(\mathbf{r}) = A_s u(\mathbf{r} - \mathbf{c}_s), \quad \phi_s = \phi_{x,i} + \phi_{y,j}, \quad H_m(\mathbf{r}) = \sum_{s: k_s=m} g_s(\mathbf{r}) e^{i\phi_s}. \quad (3)$$

Only $N_x + N_y$ of the $N_x N_y$ spot phases are free. Dropping the global factor $e^{i2\pi f_{\min} t}$,

$$\boxed{E(\mathbf{r}, t) = \sum_m H_m(\mathbf{r}) e^{i2\pi m f_0 t}, \quad I(\mathbf{r}, t) = \sum_\delta D_\delta(\mathbf{r}) e^{i2\pi \delta f_0 t},} \quad (4)$$

$$D_\delta(\mathbf{r}) = \sum_{k_s - k_{s'} = \delta} g_s(\mathbf{r}) g_{s'}(\mathbf{r}) e^{i(\phi_s - \phi_{s'})}, \quad D_{-\delta} = D_\delta^*, \quad f_{\text{beat}} = \delta f_0. \quad (5)$$

Time average over T_0 and its two parts:

$$\langle I \rangle_t = D_0 = \underbrace{\sum_s g_s^2}_{\text{incoherent}} + 2 \sum_{\substack{s < s' \\ k_s = k_{s'}}} g_s g_{s'} \cos(\phi_s - \phi_{s'}), \quad \text{static term} \xrightarrow{\phi_s - \phi_{s'} = \pm\pi/2} 0. \quad (6)$$

Temporal fluctuation — the orders are orthogonal over T_0 , so the *variances* add (Parseval):

$$\text{Var}_t[I] = \sum_{\delta \neq 0} |D_\delta|^2 = \sum_{\delta > 0} \underbrace{2|D_\delta|^2}_{=: \sigma_\delta^2}, \quad u_\delta := \left\langle \frac{\sigma_\delta^2}{\langle I \rangle_t^2} \right\rangle_R^{1/2}, \quad U_t = \left(\sum_{\delta > 0} u_\delta^2 \right)^{1/2}, \quad (7)$$

with $\langle \cdot \rangle_R$ the average over the spot square R ($2.34 \times 2.34 \mu\text{m}$), the same region as for the spatial uniformity U_h ; being linear, it may be taken inside the sum. The extreme order $\delta = k_{\max} - k_{\min}$ comes from one corner pair alone, so $|D_{\delta_{\max}}|$ is phase-independent.

3 × 4 at the working point. $W = 0.37$ MHz, $f_{\text{off}} = 100$ MHz, $k_{i,j} = 3i + 2j$, $f_0 = W/6 = 61.67$ kHz, $T_0 = 16.22$ μs; pitches $p_x = 1.169$ μm, $p_y = 0.779$ μm.

$k_{i,j}$	$j = 0$	$j = 1$	$j = 2$	$j = 3$
$i = 0$	0	2	4	6
$i = 1$	3	5	7	9
$i = 2$	6	8	10	12

δ	f_{beat} [kHz]	$(\Delta i, \Delta j)$	u_δ [%]
1	61.7	(1, -1), (-1, 2)	20.0
2	123.3	(0, 1), (2, -2)	40.5
3	185.0	(1, 0), (-1, 3)	51.7
4	246.7	(0, 2), (2, -1)	20.4
5	308.3	(1, 1)	18.8
6	370.0	(2, 0), (0, 3)	16.2
7	431.7	(1, 2)	7.0
8	493.3	(2, 1)	3.2
9	555.0	(1, 3)	6.1
10	616.7	(2, 2)	5.4
12	740.0	(2, 3)	1.8
$U_t = (\sum_\delta u_\delta^2)^{1/2}$			76.6

Evaluated by Eq. (7) for the tone phases $\phi_x = (0, 12.5, 25)^\circ$, $\phi_y = (0, 98, 16, 114)^\circ$ of Sec. 3; $\delta = 11$ is unoccupied. Weighting by the atom's position distribution instead of the spot square gives $u_\delta = 23.7, 28.8, 49.9, 32.6, 23.6, 8.1, 1.5, 4.4, 2.5, 2.1, 0.3$ and $U_t = 74.8\%$: the high orders are suppressed at the atom because they come from the most distant spot pairs. One degenerate pair: $k_{(0,3)} = k_{(2,0)} = 6$, the diagonally opposite corners, $\varrho = 3.305$ μm apart, weight $A^2 = r_x r_y$; the extreme order $\delta = 12$ comes from that same pair alone, so no phase choice can change it.

Assumptions: scalar field of common polarisation, both AODs in the same diffraction order, focal plane (real modes), commensurate tone frequencies.

3 Choice of the tone phases

3.1 The centring criterion

Requirement — the atom at $\mathbf{r} = 0$, smeared over $\sigma \approx 108$ nm [1], must see a pattern that is symmetric at *every* instant, not merely on average:

$$I(-\mathbf{r}, t) = I(\mathbf{r}, t) \quad \forall t \quad \Longleftrightarrow \quad E(-\mathbf{r}, t) = e^{i\alpha(t)} E^*(\mathbf{r}, t), \quad (8)$$

with one and the same $\alpha(t)$ for the whole field. Starting from

$$E(\mathbf{r}, t) = \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{-i2\pi(f_n + g_m)t} e^{i(\varphi_n + \psi_m)}, \quad (9)$$

three symmetries of the profile are used, with the mirror index $\bar{n} = N - 1 - n$:

$$\mathbf{c}_{\bar{n}\bar{m}} = -\mathbf{c}_{nm}, \quad (\text{centred spot grid}) \quad (10)$$

$$a_{\bar{n}} = a_n, \quad b_{\bar{m}} = b_m, \quad (\text{amplitudes } [\varrho, 1, \dots, 1, \varrho]) \quad (11)$$

$$f_{\bar{n}} = F - f_n, \quad g_{\bar{m}} = G - g_m, \quad F = f_n + f_{\bar{n}} = 2f_{\text{off}} + (N - 1)d. \quad (12)$$

Evaluating E at $-\mathbf{r}$ and relabelling the sum by the mirror index:

Step 1 (positions, u even): $u(-\mathbf{r} - \mathbf{c}_{\bar{n}\bar{m}}) = u(-(\mathbf{r} - \mathbf{c}_{nm})) = u(\mathbf{r} - \mathbf{c}_{nm})$ — the envelope factor is reproduced.

Step 2 (amplitudes): $a_{\bar{n}} b_{\bar{m}} = a_n b_m$ — unchanged.

Step 3 (frequencies), the decisive one:

$$e^{-i2\pi(f_{\bar{n}} + g_{\bar{m}})t} = e^{-i2\pi(F + G)t} \underbrace{e^{+i2\pi(f_n + g_m)t}}_{\text{conjugate of the original}}. \quad (13)$$

Step 4 (collect and compare):

$$E(-\mathbf{r}, t) = e^{-i2\pi(F+G)t} \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{+i2\pi(f_n+g_m)t} e^{i(\varphi_{\bar{n}}+\psi_{\bar{m}})}, \quad (14)$$

$$E^*(\mathbf{r}, t) = \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{+i2\pi(f_n+g_m)t} e^{-i(\varphi_n+\psi_m)}. \quad (15)$$

Same basis functions, same real coefficients: proportional iff $e^{i(\varphi_{\bar{n}}+\psi_{\bar{m}})} = e^{i\Theta} e^{-i(\varphi_n+\psi_m)}$ for all n, m with a *single* constant Θ , hence

$$\boxed{\varphi_n + \varphi_{N_x-1-n} = c_x \quad \forall n, \quad \psi_m + \psi_{N_y-1-m} = c_y \quad \forall m.} \quad (16)$$

The mirror *conjugates* the time factor, so the phases must sum to a constant; mirror-symmetric sets ($\varphi_{\bar{n}} = \varphi_n$) do not centre the pattern and leave centroid offsets of 44 nm to 84 nm against $\sigma = 108$ nm, versus 0.00 nm for Eq. (16).

3.2 Structure of the admissible family

With $\varphi_n = c/2 + \chi_n$, Eq. (16) becomes

$$\chi_{\bar{n}} = -\chi_n \quad \implies \quad \dim(\text{family}) = \left\lfloor \frac{N}{2} \right\rfloor \quad \text{instead of } N-1. \quad (17)$$

For odd N the map $n \mapsto \bar{n}$ has a fixed point ($N=3$: $0 \leftrightarrow 2$, $1 \mapsto 1$), so the middle tone is paired with *itself* and Eq. (16) pins its phase:

$$2\varphi_{(N-1)/2} = c \quad \implies \quad \varphi_{(N-1)/2} = \frac{c}{2} \quad \text{or} \quad \frac{c}{2} + \pi. \quad (18)$$

A ramp $\chi_n = \delta \hat{n}$, $\hat{n} := n - \frac{N-1}{2}$, satisfies Eq. (17) automatically and is a pure time shift of the envelope,

$$A_\delta(t) = A_0(t + \tau), \quad \tau = \frac{\delta}{2\pi d}, \quad (19)$$

hence crest-neutral and redundant with the trigger t_0 . Of the $\lfloor N/2 \rfloor$ dimensions one is therefore always inactive:

$$\dim(\text{crest-relevant}) = \left\lfloor \frac{N}{2} \right\rfloor - 1 \xrightarrow{N=3} 0. \quad (20)$$

3.3 The crest factor, and why $N_x = 3$ is locked

$$C = \frac{\max_t |s(t)|}{\text{rms}_t[s(t)]} = \sqrt{2} \frac{\max_t |A(t)|}{\sqrt{\sum_n a_n^2}}, \quad \sqrt{2} \leq C \leq \sqrt{2} \frac{\sum_n a_n}{\sqrt{\sum_n a_n^2}} \xrightarrow{a_n \equiv 1} \sqrt{2N}, \quad (21)$$

both operations over a full envelope period; the denominator is phase-independent [2, 3]. For $N=3$, inserting Eq. (16) as $\varphi = (\frac{c}{2} - \chi, \frac{c}{2} \text{ or } \frac{c}{2} + \pi, \frac{c}{2} + \chi)$ and factoring out $e^{ic/2}$, with $\theta := 2\pi dt$:

$$\boxed{A(t)e^{-ic/2} = \varrho_x e^{-i(\chi+\theta)} \pm 1 + \varrho_x e^{+i(\chi+\theta)} = 2\varrho_x \cos(\theta + \chi) \pm 1.} \quad (22)$$

The only free parameter sits *inside* the cosine, and cos sweeps $[-1, 1]$ over one period regardless of χ :

$$\max_t |A| = \max_{u \in [-1, 1]} |2\varrho_x u \pm 1| = 2\varrho_x + 1 = \sum_n a_n \quad \forall c, \chi, \text{ both branches}, \quad (23)$$

which is the triangle-inequality bound of Eq. (21) — the maximum any phase set can reach. With $r_x = 0.97$, $\varrho_x = \sqrt{r_x} = 0.9848858$:

$$C_x = \sqrt{2} \frac{2\varrho_x + 1}{\sqrt{2r_x + 1}} = \sqrt{2} \frac{2.9697716}{1.7146428} = 2.4494 \quad (\sqrt{2N} = \sqrt{6} = 2.4495). \quad (24)$$

For $N_y = 4$ there is no fixed point and Eq. (16) leaves $\psi = [0, b, c_y - b, c_y]$ with two parameters, one of which is crest-relevant. At $c_y = 114^\circ$:

b	ψ [deg]	$\max A $	C_y
38°	[0, 38, 76, 114]	4.154	2.8265 (ramp, $= \sum_n a_n$)
60°	[0, 60, 54, 114]	4.022	2.7364
98°	[0, 98, 16, 114]	3.228	2.1963
114°	[0, 114, 0, 114]	3.639	2.4761

Peak-limited amplifier [4], [5, Sec. 10.3], with $P_{\text{1dB}} = +36 \text{ dBm} = 3.98 \text{ W}$ [6]:

$$P_{\text{peak}} = C^2 P_{\text{avg}} \leq P_{\text{1dB}} \quad \Rightarrow \quad P_{\text{avg}} \leq \frac{P_{\text{1dB}}}{C^2}. \quad (25)$$

Working point. $\varphi_x = (0, 12.5, 25)^\circ$, $\psi = (0, 98, 16, 114)^\circ$; both satisfy Eq. (16) ($0+25 = 2 \cdot 12.5$, $0 + 114 = 98 + 16$).

	C	C^2	$P_{\text{avg}} \leq P_{\text{1dB}}/C^2$
x axis, $N_x = 3$ (forced by Eq. (20))	2.4494	6.00	0.66 W
y axis, $N_y = 4$, $b = 98^\circ$ (minimum in the family)	2.1963	4.82	0.83 W

4 Pulse area for a given profile power

Scale factor from the optical power P in the profile, $I = s\tilde{I}$ with $\tilde{I} = |E|^2$ in units of $u(0) = 1$:

$$P = \int \langle I \rangle_t d^2r = s \tilde{P}_{\text{prof}}, \quad \tilde{P}_{\text{prof}} = \int \langle \tilde{I} \rangle_t d^2r = \sum_s a_s P_1 + 2 \sum_{\substack{s < s' \\ k_s = k_{s'}}} A_s A_{s'} \cos(\phi_s - \phi_{s'}) O(\varrho). \quad (26)$$

Mode power and autocorrelation [7–9]:

$$P_1^G = \frac{\pi w_0^2}{2}, \quad O^G(\varrho) = P_1^G e^{-\varrho^2/2w_0^2}; \quad P_1^A = \frac{4\pi}{k^2}, \quad O^A(\varrho) = P_1^A u(\varrho), \quad (27)$$

with $k = 3.8317/(1.4830 w_0)$; the Airy autocorrelation is the mode itself, sign included. Atomic response by adiabatic elimination of the $5P$ manifold, signed matrix elements and hyperfine shifts from ARC [10, 11], far-detuned scaling as in [12]:

$$\Omega = C_{\text{rabi}} I, \quad \delta_{\text{LS}} = C_{\text{shift}} I, \quad \Gamma_{\text{sc}} = C_{\text{scatter}} I, \quad \varepsilon_{\text{LS}} = \frac{\delta_{\text{LS}}}{\Omega} \text{ (a pure number)}. \quad (28)$$

Pulse area of a rectangular pulse $[t_0, t_0 + T_p]$:

$$\theta(\mathbf{r}, t_0) = C_{\text{rabi}} P \frac{\int_{t_0}^{t_0+T_p} \tilde{I}(\mathbf{r}, t) dt}{\int \langle \tilde{I} \rangle_t d^2r}, \quad (29)$$

in which the mode normalisation cancels. Inserting Eq. (4) and integrating the window term by term,

$$\int_{t_0}^{t_0+T_p} e^{i2\pi\delta f_0 t} dt = T_p \text{sinc}(\delta f_0 T_p) e^{i2\pi\delta f_0 (t_0+T_p/2)}, \quad (30)$$

$$\theta(\mathbf{r}, t_0) = \frac{C_{\text{rabi}} P T_p}{\tilde{P}_{\text{prof}}} \left[D_0 + 2 \sum_{\delta > 0} |D_\delta| \text{sinc}(\delta f_0 T_p) \cos(2\pi\delta f_0 (t_0 + \frac{T_p}{2}) + \arg D_\delta) \right]. \quad (31)$$

Limits: $T_p \gg T_0$ damps every $\delta \neq 0$ (the pulse sees the time average); $T_p \ll T_0$ gives $\text{sinc} \rightarrow 1$ (the pulse samples the beating and θ depends on t_0). Atom-weighted quantities, with $W(\mathbf{r})$ the thermal position distribution of width $\sigma^2 = \frac{\hbar}{2m\omega_r} \coth \frac{\hbar\omega_r}{2k_B T}$ [1]:

$$c_\delta = \int W D_\delta d^2r, \quad U(\theta) = \frac{\sigma_W(\theta)}{\langle \theta \rangle_W}, \quad \langle p_\uparrow \rangle_W = \int W \sin^2 \frac{\theta}{2} d^2r \neq \sin^2 \frac{\langle \theta \rangle_W}{2}. \quad (32)$$

Since θ is linear in the D_δ , a full trigger scan is one matrix–vector product in the c_δ .

Working point. 3×4 , Airy, $w_0 = 1.04 \mu\text{m}$, $r_x/r_y = 0.97/1.16$, $\phi = 0$, $\Delta = +50 \text{ GHz}$, $P = 10 \mu\text{W}$ in the profile, $T_p = 1 \mu\text{s}$, $\sigma = 108 \text{ nm}$:

$\tilde{P}_{\text{prof}}$	$2.615 \times 10^{-11} \text{ m}^2$	$\langle \theta \rangle_W$ at best trigger	11.54π
C_{rabi}	$11.66 \text{ rad s}^{-1}/(\text{W}/\text{m}^2)$	$U(\theta)$ there	0.72%
ε_{LS}	-0.061	P for a $1 \mu\text{s}$ π pulse	$0.87 \mu\text{W}$
$A_{\text{eff}}, f_{\text{Rabi}}$	$13.67 \mu\text{m}^2, 1.358 \text{ MHz}$	rephasing gain over $\langle I \rangle_t$	4.25

Not included: static two-photon detuning and magnetic shifts, uncompensated light-shift gradients, spontaneous emission during the pulse, the Zeeman substructure, atomic motion, finite pulse edges and the acoustic fill time, polarisation and vector light shifts. P is the power *in the profile*, i.e. after both AODs.

5 Diffraction efficiency and profile power

Single grating, Bragg regime [13–15], with $\Delta n \propto \sqrt{P_{\text{ac}}}$ from the photoelastic effect:

$$\eta(P) = \alpha(\nu) \sin^2 \left(\frac{\pi}{2} \sqrt{\frac{P}{P_s(\nu)}} \right), \quad \kappa = \frac{\pi}{\lambda} \sqrt{\frac{M_2 L}{2H}} \sqrt{\zeta}, \quad P_s = \left(\frac{\pi}{2\kappa} \right)^2. \quad (33)$$

N tones share one undiffracted wave; the coupled-wave equations [16, 17] reduce exactly, giving [18, 19]

$$\eta_{\text{tot}} = \alpha \sin^2 \left(\frac{\pi}{2} \sqrt{\frac{P_{\text{tot}}}{P_s}} \right), \quad \eta_n = \frac{P_n}{P_{\text{tot}}} \eta_{\text{tot}}, \quad P_{\text{tot}} = \sum_n P_n. \quad (34)$$

The total depends only on P_{tot} , not on its distribution, and the tone phases do not enter at all. (The naive $\sum_n \sin^2(\kappa \sqrt{P_n})$ is the small-signal limit and is 14 % high at the working point.) Combining with Eq. (25):

$$\eta_{\text{axis}}(C) = \alpha \sin^2 \left(\frac{\pi}{2C} \sqrt{\frac{P_{\text{1dB}}}{P_s}} \right), \quad \eta_{\text{profile}} = \eta_x \eta_y \Xi. \quad (35)$$

Parameters measured on this device family [18]: $\alpha \approx 0.92$ per axis, $P_s \approx 1.70/1.62 \text{ W}$ for x/y near 100 MHz, and the cross-axis term $\Xi_{ij} = (1 - \xi_0(\nu_i^{(x)} - \xi_x)^2(\nu_j^{(y)} - \xi_y)^2)^2$ with $\xi_x = 108.0 \text{ MHz}$, $\xi_y = 102.3 \text{ MHz}$, falling to 0.80 in the corners of the full scan range but common to all twelve spots over the 0.37 MHz span used here.

	C	P_{avg}	η_{axis}
x axis, 3 tones, centred (forced)	2.449	0.66 W	0.636
y axis, 4 tones, $b = 98^\circ$	2.196	0.83 W	0.746
y axis, ramp (worst in the family)	2.827	0.50 W	0.520
hypothetical $C = \sqrt{2}$	1.414	1.99 W	0.905
$\eta_{\text{profile}} = \eta_x \eta_y$ at the working point			47.4 %
... at $C = \sqrt{2}$ on both axes (factor 1.71)			80.9 %
laser power for $10 \mu\text{W}$ in the profile			$21.1 \mu\text{W}$
... for the $0.87 \mu\text{W}$ of a π pulse			$1.8 \mu\text{W}$

Assumptions: common Bragg matching for all tones (the 0.37 MHz span costs 0.007 % against the 36 MHz bandwidth by the detuned two-wave solution [16]); first order only; a linear transducer — the instantaneous strain carries C as well, so a high C also brings on acoustic nonlinearity; isotropic diffraction, whereas the device is anisotropic and birefringent (qualitative agreement, 20); and a two-tone OIP3 estimate for the intermodulation, 51 dBc here, which is optimistic once many tones contribute to the same bin.

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